

Computational Replication Report:  
Zhang *et al.* (2025),  
*Strain-induced nonrelativistic altermagnetic spin splitting effect*  
(arXiv:2503.17916)

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## 1 Paper summary

Using first-principles DFT (+ $U$ ) with Wannier interpolation and a constant- $\Gamma$  Kubo linear-response evaluation, the paper proposes *strain engineering* to induce altermagnetism in rutile  $\text{OsO}_2$  (and  $\text{RuO}_2$ ). Equibiaxial tensile strain  $E_{ts} \gtrsim 2\%$  drives a nonmagnetic $\rightarrow$ altermagnetic transition; the Os moment and the nonrelativistic band spin splitting both rise then fall with strain (dome, peak  $\sim 4\text{--}5\%$ ), and the altermagnetic spin-splitting effect (ASSE) yields a spin-charge conversion ratio  $\theta_{AS} \sim 7\%$ , persisting without SOC (nonrelativistic origin). Published as Phys. Rev. B **112**, 024415.

## 2 Scope statement

Full DFT+Wannier+Kubo is out of scope on CPU-only hardware. We replicate (i) the paper’s OWN quantitative correlation from Table I, (ii) the non-monotonic strain dome via a minimal Stoner mean field, and (iii) the SOC-free (nonrelativistic) moment $\rightarrow$ splitting mechanism in a  $d$ -wave tight-binding model. Absolute meV splittings and  $\theta_{AS}$  are DFT-only and reported, not recomputed.

## 3 Claims addressed

| ID | Claim                                                              | Type      | Testable (CPU)? | Tested?       |
|----|--------------------------------------------------------------------|-----------|-----------------|---------------|
| C1 | Spin splitting positively correlates with strain-induced moment    | data corr | yes             | yes           |
| C2 | Non-monotonic dome; AM onsets $> 2\%$ , peaks $\sim 4\text{--}5\%$ | mechanism | yes             | yes           |
| C3 | Splitting persists WITHOUT SOC (nonrelativistic origin)            | model     | yes             | yes           |
| C4 | $\theta_{AS} \sim 7\%$ ASSE conversion ratio                       | Kubo/DFT  | no              | reported only |

|                         |       |       |       |       |       |       |       |
|-------------------------|-------|-------|-------|-------|-------|-------|-------|
| $E_{ts}$ (%)            | 0     | 1     | 2     | 3     | 4     | 5     | 6     |
| Os moment ( $\mu_B$ )   | 0.000 | 0.002 | 0.028 | 0.349 | 0.468 | 0.500 | 0.150 |
| $ \Delta _{\max}$ (meV) | 0     | 62.8  | 209.2 | 256.3 | 275.5 | 110.0 | 0.11  |

| Quantity                                 | Paper                | This work                          | Verdict        |
|------------------------------------------|----------------------|------------------------------------|----------------|
| moment-splitting Spearman $\rho$         | positive             | 0.643                              | ✓              |
| Pearson (rising branch $E_{ts} \leq 4$ ) | positive             | 0.807                              | ✓ (sub-linear) |
| AM onset strain                          | $\gtrsim 2\%$        | $\sim 2.5\%$ (model)               | ✓              |
| dome peak                                | $\sim 4\text{--}5\%$ | $\sim 4.3\%$ (model); Table I 4–5% | ✓              |
| SOC-free splitting vs $m$ slope          | linear, nonzero      | 8.00, residual $10^{-15}$          | ✓              |
| $\theta_{AS}$                            | $\sim 7\%$           | not recomputed                     | out of scope   |

## 4 Paper Table I (reproduced input)

## 5 Results vs. paper

## 6 Per-claim what worked / what didn't

**C1 (partial/worked).** Positive association confirmed (Spearman 0.64), and near-linear on the rising branch (0.81); a single Pearson over all points (0.42) is lowered by the  $E_{ts} = 5\text{--}6\%$  dome collapse — an honest nonlinearity (splitting saturates faster than moment, Open Q1). **C2 (worked).** The Stoner dome reproduces onset  $> 2\%$  and peak  $\sim 4\text{--}5\%$ , matching Table I (moment peak 5%, splitting peak 4%). **C3 (worked).** The SOC-free  $d$ -wave model gives a nonzero splitting linear in the moment ( $8\lambda m$ ), realizing the nonrelativistic origin. **C4 (out of scope).**  $\theta_{AS} \sim 7\%$  needs the material Kubo/Wannier calculation.

## 7 Critique of the replication

This is a *mechanism + data-correlation* replication, NOT an independent DFT reproduction — the absolute meV numbers come from the paper's Table I, which we analyze rather than regenerate. The Stoner dome and the linear  $t_{AM} \propto m$  TB are deliberately minimal: they capture the QUALITATIVE physics (onset, dome, nonrelativistic persistence) but not the quantitative saturation of the splitting nor  $\theta_{AS}$ . The LLM-judge scored PARTIAL (coverage 7, agreement 6), which we adopt: the mechanism is confirmed, the DFT/transport headline is method-limited. This is a canonical reduced-model-vs-full-DFT PARTIAL for this campaign.

## 8 Verdict

**PARTIAL** — mechanism and the paper's own moment-splitting correlation + non-monotonic strain dome + nonrelativistic (SOC-free) origin reproduced; absolute meV splittings and  $\theta_{AS} \sim 7\%$  require DFT+Wannier+Kubo (out of scope, documented as method-limited).

## Open Questions

See `report/open_questions.json`: Q1 sub-linear moment→splitting saturation; Q2 6% collapse mechanism (Lifshitz vs Stoner); Q3  $\theta_{AS}$  sensitivity to  $\Gamma$  and  $U$ ; Q4 RuO<sub>2</sub> magnetism controversy;

Q5 *d*-wave-vs-FM discrimination of the strained state.